IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-239

1. Topic (12 words max)

A Multimodal System for Automated Corn Disease Detection and Farmer-Centric Knowledge Sharing

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The global cultivation of corn underpins food security for millions, yet yields are critically threatened by foliar diseases such as Northern Leaf Blight, Gray Leaf Spot, and Common Rust[1][2][3]. Early and accurate detection of these diseases is essential to enable timely intervention and minimize crop losses; however, traditional approaches that rely on expert field surveys and laboratory-based assessments are not practical for smallholder farmers or scalable across diverse agroecological regions[4].

Recent advances in artificial intelligence (AI), particularly deep learning and computer vision, have demonstrated state-of-the-art performance in automated plant disease detection, often matching or exceeding expert-level accuracy in controlled settings[1][2][3][5]. Convolutional neural networks (CNNs), including EfficientNet, DenseNet, and ResNet, have achieved classification accuracies above 98% for corn leaf disease images[1][2][3]. However, there remain significant challenges to practical deployment. Most AI models are trained and validated on benchmark datasets with ideal imaging conditions, which do not reflect the real-world variability of farmer-captured images—often characterized by poor lighting, unusual angles, or low resolution[5]. This gap leads to decreased model reliability when such systems are deployed on-farm. Furthermore, many deep learning models function as black boxes, offering no interpretability to end-users and thereby undermining trust and practical actionability, unless augmented with explainable AI techniques such as SHAP or LIME[6].

Beyond disease detection, agronomic management depends heavily on understanding crop growth stages, which informs appropriate timing for interventions. Yet, automated growth stage recognition especially for corn remains an under-developed area in digital agriculture. Most

current solutions either lack temporal context or provide static recommendations, reducing their field relevance [7][8]. Integrating growth stage and disease status could significantly improve advisory precision.

Equally important are the social and user-experience barriers that often impede AI adoption on the farm. Digital literacy, language mismatch, and the lack of mechanisms to integrate farmer feedback into AI tools limit uptake, especially among smallholders and marginalized groups. The literature increasingly highlights the need for farmer-centered solutions—featuring multilingual virtual agronomists powered by NLP, adaptive user interfaces, and novel socio-technical approaches such as farmer-to-farmer knowledge sharing and peer feedback to foster trust and accelerate innovation diffusion.

Addressing these intertwined problems, the proposed research aims to develop a multi-component, AI-augmented corn disease management platform. This platform will combine robust and explainable AI-based disease detection for field-acquired images, automated growth stage recognition for context-driven advice, a multilingual, accessible virtual agronomist and farmer-to-farmer case-based support and feedback. Together, these components respond to well-documented technical, social, and adoption challenges, advancing both the reliability and inclusivity of digital solutions in smart agriculture.

[1] H. Amin, A. Darwish, A. E. Hassanien and M. Soliman, "End-to-End Deep Learning Model for Corn Leaf Disease Classification," in IEEE Access, vol. 10, pp. 31103-31115, 2022, doi: 10.1109/ACCESS.2022.3159678.

[2] Rajeena P. P., F., S. U., A., Moustafa, M. A., & Ali, M. A. S. (2023). Detecting Plant Disease in Corn Leaf Using EfficientNet Architecture—An Analytical Approach. *Electronics*, *12*(8), 1938.

[3] Fraiwan, M., Faouri, E., & Khasawneh, N. (2022). Classification of Corn Diseases from Leaf Images Using Deep Transfer Learning. *Plants*, *11*(20), 2668.

[4] Jafar, A., Bibi, N., Naqvi, R. A., Sadeghi-Niaraki, A., & Jeong, D. (2024). Revolutionizing agriculture with artificial intelligence: plant disease detection methods, applications, and their limitations. F*rontiers in plant science, 15,* 1356260.

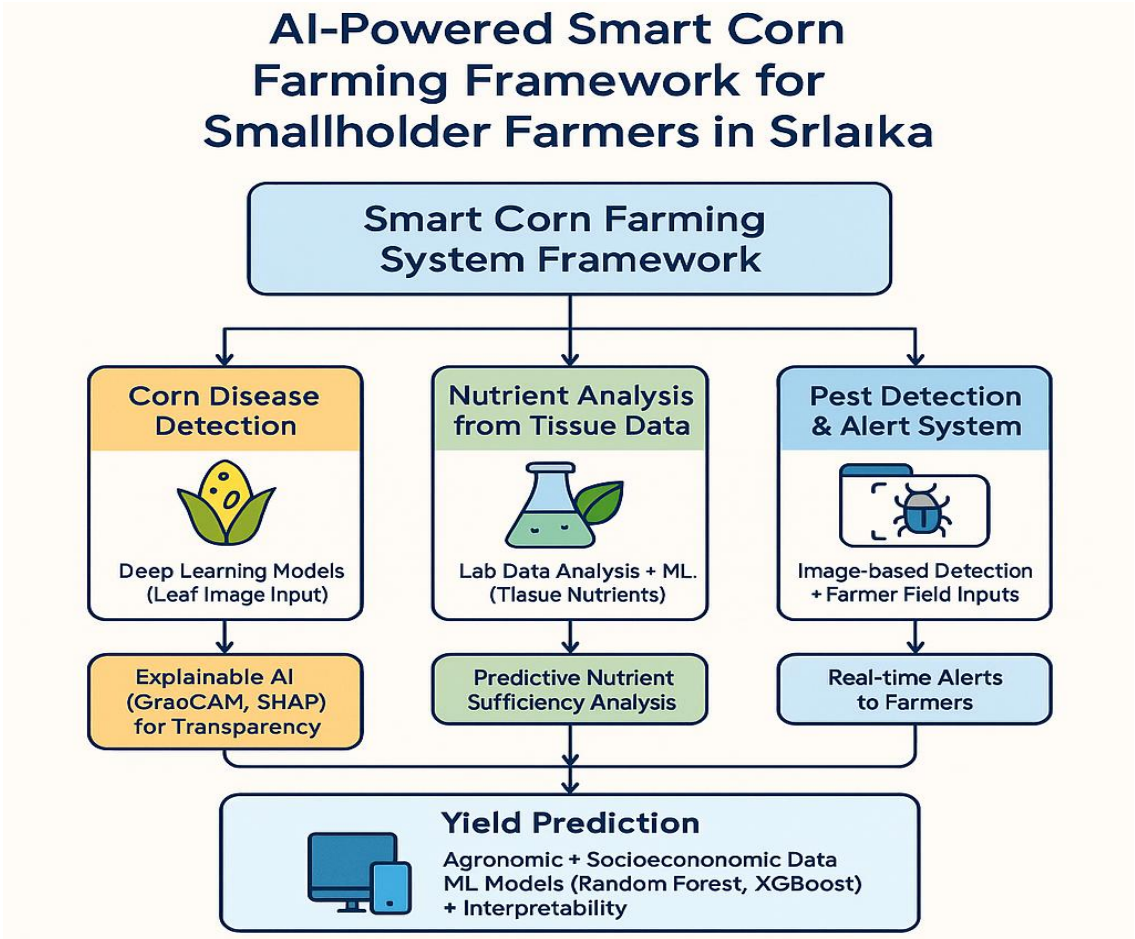
[5] Shoaib, M., Sadeghi-Niaraki, A., Ali, F., Hussain, I., & Khalid, S. (2025). Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions. F*rontiers in plant science, 16,* 1538163.

[6] A. Oad et al., "Plant Leaf Disease Detection Using Ensemble Learning and Explainable AI," in IEEE Access, vol. 12, pp. 156038-156049, 2024, doi: 10.1109/ACCESS.2024.3484574.

[7] Darwin, B., Dharmaraj, P., Prince, S., Popescu, D. E., & Hemanth, D. J. (2021). Recognition of Bloom/Yield in Crop Images Using Deep Learning Models for Smart Agriculture: A Review. A*gronomy, 11(*4), 646.

[8] Wang, Y.-H., & Su, W.-H. (2022). Convolutional Neural Networks in Computer Vision for Grain Crop Phenotyping: A Review. *Agronomy*, *12*(11), 2659.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research leverages the specialized domain of Artificial Intelligence and Machine Learning for Smart Agriculture, focusing on improving corn production through advanced data-driven solutions. The work combines core expertise in computer vision, deep learning, predictive modeling, and explainable AI, underpinned by practical understanding of agronomic and socioeconomic factors that influence smallholder farming.

Key knowledge areas include designing robust image recognition pipelines for detecting corn leaf diseases and pests under real-world field conditions, developing interpretable models that communicate insights transparently to farmers, and integrating multivariate analysis to understand nutrient deficiencies directly from tissue samples. The yield prediction component extends this expertise by combining diverse data types — such as farmer demographics, agronomic practices, and environmental factors — to deliver reliable, context-specific forecasts.

The project requires diverse datasets, including high-quality corn leaf images for training disease and pest detection models, tissue nutrient concentration data linked with field management details, and comprehensive farm-level records covering yield history, fertilizer use, advisory sources, and socioeconomic indicators. Data collection may involve partnerships with agricultural research stations, local farmer networks, and crowdsourced field reporting.

By combining advanced AI methods with locally relevant data and participatory inputs, the research aims to produce deployable, farmer-friendly tools that address key challenges in early detection, timely intervention, and sustainable yield improvement. This aligns with the specialized goal of creating explainable, accessible, and context-aware AI systems that bridge the gap between modern technology and practical smallholder farming needs.

8. Objectives and Novelty

Main Objective

To design and evaluate an integrated, data-driven framework that leverages advanced machine learning, computer vision, and explainable AI techniques to enhance corn productivity and resilience through early disease detection, pest monitoring, precise nutrient management, and context-aware yield prediction tailored for smallholder Corn farming systems.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22180452 – CHANDRASENA P.D.T.S	Develop models to accurately detect corn diseases from leaf images.	Collect and preprocess corn leaf images. Train and compare multiple deep learning models. Apply explainability techniques. Optimize models for low-resource devices.	Integrates explainable AI methods for transparent disease recognition.
IT22055958 – Perera U. S. A	Pest Detection and Alert System	Investigate and select suitable techniques for automated pest detection using computer vision or multimodal data sources. Develop and experiment with image classification or object detection models for identifying key pest species or damage symptoms.	Hybrid pest detection approach that combines automated image-based recognition with crowdsourced farmer input to improve accuracy

		Explore integration of additional farmer-reported field observations to enhance detection accuracy. Design and assess an effective alerting mechanism that communicates detection results in a timely, user-friendly manner. Evaluate system performance in terms of detection accuracy, false positives/negatives, and practical deployment feasibility.	
IT22542274 – Chathumadura K.K.K	Nutrient Composition Prediction and Analysis from Corn Tissue Samples	Conduct a comprehensive assessment of nutrient profiles (e.g., nitrogen, phosphorus, potassium, zinc) across collected tissue samples. Identify and interpret patterns and anomalies in nutrient levels under varying farming practices and site conditions. Formulate predictive approaches for estimating nutrient sufficiency or deficiency based on field management data. Explore interrelationships among multiple nutrients to reveal co-deficiency or antagonistic effects. Develop practical recommendations for nutrient interventions informed by the analytical findings.	This component moves beyond standard lab analysis by connecting tissue nutrient trends with planting and environmental variables, enabling proactive nutrient management strategies tailored to field-specific conditions. It contributes new insights into how real-time tissue monitoring can inform precision fertilization and sustainable crop nutrition.

IT22600066 – Jayalathge B A	ML Based Corn Yield Prediction for Sri Lankan Farmers	Analyze and interpret relevant data to identify patterns, trends, and influential factors affecting corn yield. Design and compare suitable predictive modeling techniques for yield estimation, exploring both linear and non-linear approaches. Investigate the relative impact of socioeconomic and agronomic variables on model accuracy and reliability. Experiment with model validation strategies to ensure robustness and mitigate overfitting and analyze the implications for practical deployment.	Uniquely combines agronomic and socioeconomic data to build context-aware, interpretable yield prediction model

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22180452 – CHANDRASENA P.D.T.S	This component applies advanced computer vision and deep learning expertise to train, test, and compare multiple models for reliable disease identification. It uses explainable AI methods to make model results transparent, showing deep knowledge of both machine learning techniques and agricultural disease diagnosis.
IT22055958 – Perera U. S. A	This component demonstrates the specialization's strength in combining computer vision, machine learning, and participatory sensing for real-time agricultural intelligence. The hybrid approach of automated pest detection enhanced by farmer input ensures robust detection under diverse field conditions. The development of a practical alert system translates technical outputs into actionable knowledge, promoting timely pest management and yield protection.
IT22542274 – Chathumadura K.K.K	This part of the research leverages multivariate analysis and predictive modeling to support sustainable nutrient management, reflecting the specialization's emphasis on data-driven decision-making for crop productivity. By connecting tissue nutrient profiles with field variables and translating findings into actionable insights, it empowers precision fertilization strategies that optimize resource use and promote environmentally responsible farming practices.
IT22600066 – Jayalathge B A	This module directly applies core machine learning principles to agricultural yield forecasting, integrating both socioeconomic and agronomic variables to create context-specific models for Sri Lankan farmers. By emphasizing interpretability and model reliability, it bridges the gap between advanced predictive analytics and practical advisory services, showcasing how specialized ML techniques can improve planning, risk management, and policy support in the local farming sector.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Harinda	Fernando	
Co-Supervisor	Mr.	Amila	Alexander	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.